

Prasanjit Dubey

Ph.D. Candidate, Machine Learning, Georgia Tech · Graduating May 2027

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Education

Georgia Institute of Technology

Expected May 2027

H. Milton Stewart School of Industrial and Systems Engineering (ISyE)

Ph.D. in Machine Learning, Minor in Mathematics | Advisor: Prof. Xiaoming Huo | GPA 3.85/4.0

Indian Institute of Technology Bombay

2017 – 2019

M.Sc. in Applied Statistics & Informatics | Ranked 1st of 38 | GPA 8.96/10

Thesis: Financial Data Mining Using Multivariate Copulas

University of Calcutta

2014 – 2017

B.Sc. (Honors) in Statistics, Minor in Mathematics & Computer Science | Grade 92.5%

Professional Experience

Bloomberg L.P., New York, USA

Jun 2025 – Aug 2025

Quant Intern, Quantitative Risk Analytics (Enterprise Products)

- Built a **Kalman-filter hazard-curve model** for pricing credit instruments by estimating dynamic CDS term structures from **latent Nelson-Siegel factors and macroeconomic covariates**; **integrated the model into Bloomberg Terminal's Credit VaR engine** to forward-simulate risk-neutral hazard rates.
- Implemented a **Credit Rating Migration module** for the Credit VaR engine using **macro-conditioned multinomial logit and intensity-based models** to simulate obligor- and index-level rating paths, capturing systematic and idiosyncratic risk.
- Designed a **hybrid correlation model** combining economic factor loadings with **PCA-calibrated residuals** to simulate joint firm-value shocks across obligors.

AT&T Labs, Inc., New Jersey, USA

May 2024 – Aug 2024

Research Scientist Intern, Data Science & AI Research

- Developed a novel framework for analyzing **sparse longitudinal customer-journey text** by integrating **large language models (LLMs)** with **multivariate functional data analysis**.
- Applied **sparse multivariate functional PCA** with user covariates for segmentation, **anomaly detection**, and recovery of key modes of variation in the customer journey.
- Used **LLM-guided dynamic keyword profiling** on real-world data (Amazon reviews) to surface customer pain points and actionable insights, **outperforming existing methods for customer-journey analytics**; shipped a **production application** (Emotional AI) for behavioral anomaly detection and targeted interventions.

Australia and New Zealand Banking Group (ANZ), Bengaluru, India

Jun 2019 – Aug 2021

Senior Quantitative Analyst, Wholesale Credit Risk Modeling

- Built the **Financial Ratio Surveillance Tool (FRST)**, a production **R Shiny** stress-testing application designed to forecast default probabilities and assess the predictive power of financial ratios.
- Developed a **multivariate financial-ratio analytics suite** (sigmoid calibration, kernel regression, shape-constrained additive models, spline smoothing) paired with a novel **branched forward-selection** routine to automatically select the top- n credit-risk models across diverse portfolios.
- Led the bank's **statistical COVID-19 impact assessment** across commercial portfolios, flagging high-volatility sectors for stakeholders as part of **APRA regulatory** requirements.

Walmart Labs, Bengaluru, India

May 2018 – Aug 2018

Research Intern, Data Science (International Pricing & Demand Forecasting)

- Devised an **ensemble of ridge, lasso, and elastic-net models** using high-dimensional, large-scale data to forecast sales across product categories based on item-, region-, and covariate-specific **price elasticities**.
- Built a **cluster-based (k -means) price-elasticity model**, **reducing mean absolute percentage error (MAPE) by 42.4% relative to Walmart's incumbent methods**.

Research Papers

- [1] **Anytime-Valid Federated Conformal RAG for LLM Swarms.**
Prasanjit Dubey & Xiaoming Huo.
Under review at the Conference on Neural Information Processing Systems (NeurIPS), 2026. [Paper]
- [2] **Matching Rates and Optimal Allocation for Federated Probe-Logit Distillation under Heterogeneous Bandwidth Budgets.**
Prasanjit Dubey & Xiaoming Huo.
Invited submission under review in the Theory and Methods section of the Journal of the American Statistical Association (JASA), 2026. [Paper]
* Poster presented at Statistics and Trustworthy AI for Cross (X)-Domain Acceleration (STAI-X), Harvard University, 2026
- [3] **Federated Language Models Under Bandwidth Budgets: Distillation Rates and Conformal Coverage.**
Prasanjit Dubey & Xiaoming Huo.
Under review at the Conference on Neural Information Processing Systems (NeurIPS), 2026. [Paper]
- [4] **BOOST: Power-Optimal Strong-FWER Testing for Block-Structured Multiplicity.**
Prasanjit Dubey & Xiaoming Huo.
Under review at the Conference on Neural Information Processing Systems (NeurIPS), 2026. [Paper]
- [5] **Optimal multiple testing under family-wise error control: elementary symmetric polynomials and a scalable algorithm.**
Prasanjit Dubey & Xiaoming Huo.
Under review at Biometrika, 2026. [Paper]
- [6] **LLmFPCA-detect: LLM-powered Multivariate Functional PCA for Anomaly Detection in Sparse Longitudinal Texts.**
Prasanjit Dubey, Aritra Guha, Zhengyi Zhou, Qiong Wu, Xiaoming Huo, & Paromita Dubey.
Under review at IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2025. [Paper]
* 2nd Place Paper Award, Student Research Showcase, 2025 INFORMS Regional Analytics Conference (IRAC)
- [7] **Most Powerful Test with Exact Family-Wise Error Rate Control: Necessary Conditions and a Path to Fast Computing.**
Prasanjit Dubey & Xiaoming Huo.
Under review at the Journal of Computational and Graphical Statistics (JCGS), 2025. [Paper]
* Invited talk, Optimisation for Statistics and Learning session, 2025 INFORMS Annual Meeting
- [8] **Generalization Bounds for Magnitude-Based Pruning via Sparse Matrix Sketching.**
Etash Kumar Guha, Prasanjit Dubey, & Xiaoming Huo.
Poster, ICLR 2024 Workshop on Bridging the Gap Between Practice and Theory in Deep Learning. [Paper]

Technical Skills

Programming:	Python, R, R Shiny, C++, SQL, MySQL, MATLAB
Machine Learning:	LLMs, PyTorch, scikit-learn
Computing & Tools:	HPC (Slurm), GPU computing, Apache Hadoop (MapReduce, Streaming), Git
Graduate Coursework:	High-Dimensional Probability, High-Dimensional Statistics, Statistical Inference, Time Series Analysis, Stochastic Processes, Statistical Techniques in Data Mining, Convex Optimization, Stochastic Optimization, Computational Data Analysis

Awards & Honors

2nd Place Paper Award , Student Research Showcase, INFORMS Regional Analytics Conference (IRAC), Atlanta	2025
Stewart Topper Fellowship Award , Georgia Tech	2021 – 2022
Dr. P. V. Sukhatme Memorial Award (M.Sc. rank 1), IIT Bombay	2019
All India Rank 26 , Graduate Aptitude Test in Engineering (GATE), Statistics	2019
Pfizer Scholarship (M.Sc. first-year rank 1)	2018
All India Rank 7 , IIT JAM (Mathematical Statistics)	2017
Swami Asaktananda Award (highest B.Sc. academic distinction)	2017
Summer Research Fellowship , Indian Academy of Sciences	2016, 2017

Academic Service & Teaching

Reviewer:	ICLR, ICML, and NeurIPS	2023 – present
Teaching Assistant:	> Data Mining and Statistical Learning (graduate-level, 300+ students)	2022 – 2025
(Georgia Tech)	> Statistical Modeling and Regression Analysis (graduate-level, 300+ students)	
Students Mentored:	> Kareena Mehta (Georgia Tech, CS)	
	> Kevin Li (Washington University in St. Louis, Finance and Data Science)	